

# MultiOmicsSurvey Function Documentation

Auto-generated

2025-12-16

## Contents

|          |                                                        |          |
|----------|--------------------------------------------------------|----------|
| <b>1</b> | <b>Introduction</b>                                    | <b>5</b> |
| <b>2</b> | <b>custom_functions.R</b>                              | <b>5</b> |
| 2.1      | as.sunburstDF . . . . .                                | 5        |
| 2.2      | calculate_ari_index . . . . .                          | 6        |
| 2.3      | calculate_nmi_index . . . . .                          | 6        |
| 2.4      | calculate_S . . . . .                                  | 7        |
| 2.5      | CIMLR_mod . . . . .                                    | 7        |
| 2.6      | CIMLR_Estimate_Number_of_Clusters_weight_mod . . . . . | 8        |
| 2.7      | CIMLR.weight_mod . . . . .                             | 8        |
| 2.8      | coca_cc_mod . . . . .                                  | 9        |
| 2.9      | compute_matrix_similarity . . . . .                    | 9        |
| 2.10     | compute_silhouette . . . . .                           | 10       |
| 2.11     | concordanceNetworkNMI_ANF . . . . .                    | 10       |
| 2.12     | create_annot_barchart . . . . .                        | 10       |
| 2.13     | create_density_curve . . . . .                         | 11       |
| 2.14     | create_density_plot_color . . . . .                    | 11       |
| 2.15     | create_MO_heatmap . . . . .                            | 11       |
| 2.16     | dist2.cimlr_mod2 . . . . .                             | 12       |
| 2.17     | dist2.cimlr.weight_mod . . . . .                       | 12       |
| 2.18     | divide_by_quantile_normalize . . . . .                 | 13       |
| 2.19     | estimateNumberOfClustersGivenGraph_mod . . . . .       | 13       |
| 2.20     | fetch_citation . . . . .                               | 13       |
| 2.21     | fetch_in_a_nutshell . . . . .                          | 14       |
| 2.22     | frobenius_norm . . . . .                               | 14       |
| 2.23     | getMoHeatmap_prenorm_quart . . . . .                   | 14       |
| 2.24     | linear_quantile_normalize . . . . .                    | 15       |

|          |                                                |           |
|----------|------------------------------------------------|-----------|
| 2.25     | mds_from_original_matrix . . . . .             | 15        |
| 2.26     | MOVICS_jaccard_index . . . . .                 | 16        |
| 2.27     | multiple.kernel.cimlr_mod . . . . .            | 16        |
| 2.28     | multiple.kernel.cimlr.weight_mod . . . . .     | 16        |
| 2.29     | nemo.affinity.graph_mod . . . . .              | 17        |
| 2.30     | normalize_affinity_matrix . . . . .            | 17        |
| 2.31     | pca_from_original_matrix . . . . .             | 18        |
| 2.32     | pca_from_sim_matrix . . . . .                  | 18        |
| 2.33     | pearson_correlation . . . . .                  | 19        |
| 2.34     | plot_MOFA_cumul_var_exp . . . . .              | 19        |
| 2.35     | plot_MOFA_var_exp . . . . .                    | 19        |
| 2.36     | rankFeaturesByNMI_parallelly . . . . .         | 20        |
| 2.37     | rankFeaturesByNMI_parallelly_ANF . . . . .     | 20        |
| 2.38     | reshape_Pearson_for_tests . . . . .            | 20        |
| 2.39     | standardize_rows . . . . .                     | 21        |
| 2.40     | TCGAanalyze_survival_custom3 . . . . .         | 21        |
| <b>3</b> | <b>fast_pathfindR_hclust.R</b>                 | <b>22</b> |
| 3.1      | cluster_enriched_terms_fast . . . . .          | 22        |
| 3.2      | cluster_graph_vis_fast . . . . .               | 22        |
| 3.3      | create_kappa_matrix_fast . . . . .             | 22        |
| 3.4      | fuzzy_term_clustering_fast . . . . .           | 23        |
| 3.5      | hierarchical_term_clustering_fast . . . . .    | 23        |
| <b>4</b> | <b>modified_MOVICS_functions.R</b>             | <b>24</b> |
| 4.1      | compAgree2 . . . . .                           | 24        |
| 4.2      | compAgree_single_algorithm . . . . .           | 24        |
| 4.3      | compClinvar2 . . . . .                         | 25        |
| 4.4      | compClinvar_ordinal_single_algorithm . . . . . | 25        |
| 4.5      | compClinvar_single_algorithm . . . . .         | 25        |
| 4.6      | compDrugsen_single_algorithm . . . . .         | 26        |
| 4.7      | compFGA_mod . . . . .                          | 26        |
| 4.8      | compFGA_optimized . . . . .                    | 27        |
| 4.9      | compMut_single_algorithm . . . . .             | 27        |
| 4.10     | compMut_single_algorithm_sc . . . . .          | 27        |
| 4.11     | compSurv_ext . . . . .                         | 28        |
| 4.12     | getMoHeatmap_single_algorithm . . . . .        | 28        |

|          |                                                |           |
|----------|------------------------------------------------|-----------|
| 4.13     | getMoHeatmap_single_algorithm2 . . . . .       | 28        |
| 4.14     | getSilhouette_ggplot . . . . .                 | 29        |
| 4.15     | ntp_mod . . . . .                              | 29        |
| 4.16     | plot_pathway_heatmaps . . . . .                | 30        |
| 4.17     | prepare_gsea_output_for_hclust . . . . .       | 30        |
| 4.18     | runDEA_mod . . . . .                           | 30        |
| 4.19     | runGSEA_mod . . . . .                          | 31        |
| 4.20     | runGSEA_mod_4.4 . . . . .                      | 31        |
| 4.21     | runGSEA_mod_4.4_single_algorithm . . . . .     | 32        |
| 4.22     | runGSVA_mod_4.4 . . . . .                      | 32        |
| 4.23     | runGSVA_mod_4.4_single_algorithm . . . . .     | 32        |
| 4.24     | runKappa_single_algorithm . . . . .            | 33        |
| 4.25     | runMarker_mod_4.4 . . . . .                    | 33        |
| 4.26     | runMarker_single_algorithm . . . . .           | 33        |
| 4.27     | runMarker_single_algorithm_no_export . . . . . | 34        |
| 4.28     | runNTP_mod . . . . .                           | 34        |
| 4.29     | runPAM_single_algorithm . . . . .              | 34        |
| 4.30     | subHeatmap_mod . . . . .                       | 35        |
| 4.31     | twoclassdeseq2_mod . . . . .                   | 35        |
| 4.32     | twoclassedger_mod . . . . .                    | 35        |
| 4.33     | twoclasslimma_mod . . . . .                    | 36        |
| <b>5</b> | <b>Spectrum_functions.R</b>                    | <b>36</b> |
| 5.1      | CNN_kernel_mod . . . . .                       | 36        |
| 5.2      | kernfinder_local_mod . . . . .                 | 37        |
| 5.3      | kernfinder_mine_mod . . . . .                  | 37        |
| 5.4      | rbfkernel_b_mod . . . . .                      | 37        |
| 5.5      | Spectrum_bin_and_par . . . . .                 | 38        |
| <b>6</b> | <b>single_algorithm/ Helper Functions</b>      | <b>39</b> |
| 6.1      | MAD-Based Weighting Functions . . . . .        | 39        |
| 6.1.1    | mad_based_weights . . . . .                    | 39        |
| 6.1.2    | weighted_euclidean_dist . . . . .              | 39        |
| 6.1.3    | weighted_hamming_fraction . . . . .            | 39        |
| 6.1.4    | weighted_mutations . . . . .                   | 40        |
| 6.2      | Matrix Similarity Functions . . . . .          | 40        |
| 6.2.1    | compute_matrix_similarity . . . . .            | 40        |

|          |                                                               |           |
|----------|---------------------------------------------------------------|-----------|
| 6.3      | Spectral Clustering Functions . . . . .                       | 41        |
| 6.3.1    | spectralClustering_eig . . . . .                              | 41        |
| 6.3.2    | spectral_clustering_eig . . . . .                             | 41        |
| 6.3.3    | extract_eigenspace . . . . .                                  | 42        |
| 6.4      | Statistical Analysis Functions . . . . .                      | 42        |
| 6.4.1    | unbiased.cv.test . . . . .                                    | 42        |
| 6.5      | Graph/Network Analysis Functions . . . . .                    | 43        |
| 6.5.1    | determine_edge_support . . . . .                              | 43        |
| 6.6      | MONET-Specific Functions . . . . .                            | 43        |
| 6.6.1    | offset_correlation . . . . .                                  | 43        |
| 6.6.2    | edge_absolute_mean . . . . .                                  | 44        |
| 6.6.3    | get_mod_edge_absmean . . . . .                                | 44        |
| 6.7      | COCA-Specific Functions . . . . .                             | 44        |
| 6.7.1    | hclust_gap . . . . .                                          | 44        |
| 6.8      | PAMOGK Preprocessing Functions . . . . .                      | 45        |
| 6.8.1    | clean_rownames . . . . .                                      | 45        |
| 6.8.2    | remove_na_rownames . . . . .                                  | 45        |
| 6.8.3    | get_dimensions . . . . .                                      | 45        |
| 6.8.4    | fix_uniprot_row_names . . . . .                               | 45        |
| 6.8.5    | filter_uniprot_ids . . . . .                                  | 46        |
| 6.9      | iClusterBayes Tuning Functions . . . . .                      | 46        |
| 6.9.1    | prepare_global_data . . . . .                                 | 46        |
| 6.9.2    | prepare_modality_data . . . . .                               | 46        |
| 6.9.3    | get_summary_statistics_text . . . . .                         | 47        |
| 6.9.4    | create_global_plot . . . . .                                  | 47        |
| 6.9.5    | create_modality_plot . . . . .                                | 47        |
| 6.9.6    | compute_robust_metrics_and_score . . . . .                    | 47        |
| 6.9.7    | penalize_acceptance . . . . .                                 | 47        |
| 6.10     | HPC Folder Functions . . . . .                                | 48        |
| 6.10.1   | LRACluster_HPC Functions (full_LRACluster_source.R) . . . . . | 48        |
| 6.10.2   | RWR-F_HPC Functions (RWR-F_source.R) . . . . .                | 50        |
| <b>7</b> | <b>Scripts/Consensus Functions</b>                            | <b>51</b> |
| 7.1      | calc_ari . . . . .                                            | 51        |

|                                                             |           |
|-------------------------------------------------------------|-----------|
| <b>8 Scripts/MOVICS Functions</b>                           | <b>52</b> |
| 8.1 check_missing_values_with_indices . . . . .             | 52        |
| 8.2 preprocessText . . . . .                                | 52        |
| 8.3 overlap_coefficient . . . . .                           | 52        |
| 8.4 calculate_similarities . . . . .                        | 53        |
| <b>9 Scripts/method_comparisons/Comparisons.R Functions</b> | <b>53</b> |
| 9.1 calculate_pathway_similarity_matrix . . . . .           | 53        |
| 9.2 calculate_ari_pathway . . . . .                         | 54        |
| 9.3 cell_fun (inline) . . . . .                             | 54        |
| 9.4 pca_from_ari_matrix . . . . .                           | 54        |
| <b>10 Top-level Scripts Functions</b>                       | <b>55</b> |
| 10.1 check_duplicate_rownames . . . . .                     | 55        |
| 10.2 filter_most_variant_rows . . . . .                     | 55        |
| <b>11 Alphabetical Index</b>                                | <b>57</b> |

# 1 Introduction

This document provides comprehensive documentation for all custom functions used in the MultiOmicsSurvey project. Functions are organized by source file and presented in alphabetical order within each section.

## 2 custom\_functions.R

Functions for multi-omics data analysis, clustering evaluation, visualization, and data processing.

### 2.1 as.sunburstDF

```
as.sunburstDF(DF, value_column = NULL, add_root = FALSE)
```

**Description:** Converts a hierarchical data frame into a sunburst-compatible format for visualization.

**Parameters:**

| Parameter    | Type       | Description                                   |
|--------------|------------|-----------------------------------------------|
| DF           | data.frame | Hierarchical data frame with category columns |
| value_column | character  | Column name containing values for sizing      |

| Parameter | Type    | Description                                 |
|-----------|---------|---------------------------------------------|
| add_root  | logical | Whether to add a root node (default: FALSE) |

**Returns:** A data frame formatted for sunburst charts with `ids`, `labels`, `parents`, and `values` columns.

## 2.2 calculate\_ari\_index

```
calculate_ari_index(cluster_df1, cluster_df2,
                    cluster_col1 = "Cluster", cluster_col2 = "Cluster",
                    sample_col1 = "samID", sample_col2 = "samID")
```

**Description:** Calculates the Adjusted Rand Index (ARI) between two clustering results.

**Parameters:**

| Parameter    | Type       | Description                                          |
|--------------|------------|------------------------------------------------------|
| cluster_df1  | data.frame | First clustering result                              |
| cluster_df2  | data.frame | Second clustering result                             |
| cluster_col1 | character  | Column name for clusters in df1 (default: "Cluster") |
| cluster_col2 | character  | Column name for clusters in df2 (default: "Cluster") |
| sample_col1  | character  | Column name for sample IDs in df1 (default: "samID") |
| sample_col2  | character  | Column name for sample IDs in df2 (default: "samID") |

**Returns:** Numeric value representing the Adjusted Rand Index (-1 to 1).

## 2.3 calculate\_nmi\_index

```
calculate_nmi_index(cluster_df1, cluster_df2,
                    cluster_col1 = "Cluster", cluster_col2 = "Cluster",
                    sample_col1 = "samID", sample_col2 = "samID")
```

**Description:** Calculates the Normalized Mutual Information (NMI) between two clustering results.

**Parameters:**

| Parameter   | Type       | Description             |
|-------------|------------|-------------------------|
| cluster_df1 | data.frame | First clustering result |

| Parameter                 | Type                    | Description                                          |
|---------------------------|-------------------------|------------------------------------------------------|
| <code>cluster_df2</code>  | <code>data.frame</code> | Second clustering result                             |
| <code>cluster_col1</code> | <code>character</code>  | Column name for clusters in df1 (default: "Cluster") |
| <code>cluster_col2</code> | <code>character</code>  | Column name for clusters in df2 (default: "Cluster") |
| <code>sample_col1</code>  | <code>character</code>  | Column name for sample IDs in df1 (default: "samID") |
| <code>sample_col2</code>  | <code>character</code>  | Column name for sample IDs in df2 (default: "samID") |

**Returns:** Numeric value representing NMI (0 to 1).

## 2.4 calculate\_S

`calculate_S(W)`

**Description:** Calculates the S matrix from an affinity matrix W using the SNF transition probability formula.

**Parameters:**

| Parameter      | Type                | Description                |
|----------------|---------------------|----------------------------|
| <code>W</code> | <code>matrix</code> | Affinity/similarity matrix |

**Returns:** Normalized transition probability matrix S.

## 2.5 CIMLR\_mod

`CIMLR_mod(X, c, no.dim = NA, k = 10, cores.ratio = 1, binary_flags = NULL, binary_distance = "binary", max_iter = 100, use_rsvd = FALSE)`

**Description:** Modified CIMLR (Cancer Integration via Multi-kernel Learning) algorithm supporting mixed data types.

**Parameters:**

| Parameter           | Type                 | Description                                       |
|---------------------|----------------------|---------------------------------------------------|
| <code>X</code>      | <code>list</code>    | List of data matrices (features $\times$ samples) |
| <code>c</code>      | <code>integer</code> | Number of clusters                                |
| <code>no.dim</code> | <code>integer</code> | Number of dimensions for embedding (default: NA)  |

| Parameter       | Type      | Description                                         |
|-----------------|-----------|-----------------------------------------------------|
| k               | integer   | Number of neighbors (default: 10)                   |
| cores.ratio     | numeric   | Ratio of cores for parallel processing (default: 1) |
| binary_flags    | character | Vector indicating binary data types                 |
| binary_distance | character | Distance metric for binary data (default: "binary") |
| max_iter        | integer   | Maximum iterations (default: 100)                   |
| use_rsvd        | logical   | Use randomized SVD (default: FALSE)                 |

**Returns:** List containing cluster assignments, similarity matrix, and convergence info.

## 2.6 CIMLR\_Estimate\_Number\_of\_Clusters\_weight\_mod

```
CIMLR_Estimate_Number_of_Clusters_weight_mod(all_data, NUMC = 2:5,
                                              cores.ratio = 0, weight,
                                              binary_flags = NULL, binary_distance = NULL)
```

**Description:** Estimates optimal number of clusters for weighted CIMLR using eigengap and rotation cost.

**Parameters:**

| Parameter       | Type      | Description                                     |
|-----------------|-----------|-------------------------------------------------|
| all_data        | list      | List of data matrices                           |
| NUMC            | integer   | Range of cluster numbers to test (default: 2:5) |
| cores.ratio     | numeric   | Ratio of cores for parallel processing          |
| weight          | numeric   | Feature weights                                 |
| binary_flags    | character | Vector indicating binary data types             |
| binary_distance | character | Distance metric for binary data                 |

**Returns:** List with optimal K estimates from eigengap and rotation methods.

## 2.7 CIMLR.weight\_mod

```
CIMLR.weight_mod(X, c, no.dim = NA, k = 10, cores.ratio = 0, weight,
                 binary_flags = NULL, binary_distance = NULL, max_iter = 100)
```

**Description:** Weighted CIMLR implementation with feature importance weights.

**Parameters:**



| Parameter                            | Type    | Description           |
|--------------------------------------|---------|-----------------------|
| X                                    | list    | List of data matrices |
| c                                    | integer | Number of clusters    |
| weight                               | numeric | Feature weights       |
| (other parameters same as CIMLR_mod) |         |                       |

**Returns:** Clustering result with weighted kernel integration.

## 2.8 coca\_cc\_mod

```
coca_cc_mod(data = NULL, K = 2, B = 100, pItem = 0.8, clMethod = "hclust",
            innerLinkage = "complete", maxK = NULL)
```

**Description:** Modified COCA (Cluster of Clusters Analysis) consensus clustering.

**Parameters:**

| Parameter    | Type      | Description                                               |
|--------------|-----------|-----------------------------------------------------------|
| data         | list      | List of clustering matrices                               |
| K            | integer   | Number of final clusters (default: 2)                     |
| B            | integer   | Number of bootstrap iterations (default: 100)             |
| pItem        | numeric   | Proportion of items to sample (default: 0.8)              |
| clMethod     | character | Clustering method (default: "hclust")                     |
| innerLinkage | character | Linkage for hierarchical clustering (default: "complete") |
| maxK         | integer   | Maximum K for consensus matrix                            |

**Returns:** Consensus clustering result with cluster assignments.

## 2.9 compute\_matrix\_similarity

```
compute_matrix_similarity(matrices)
```

**Description:** Computes pairwise similarity between multiple matrices using Frobenius norm and Pearson correlation.

**Parameters:**

| Parameter | Type | Description                       |
|-----------|------|-----------------------------------|
| matrices  | list | Named list of matrices to compare |

**Returns:** List with Frobenius and Pearson similarity matrices.

## 2.10 compute\_silhouette

```
compute_silhouette(cluster_df, similarity_matrix, normalize_matrix = FALSE)
```

**Description:** Computes silhouette scores from a similarity matrix and cluster assignments.

**Parameters:**

| Parameter         | Type       | Description                                        |
|-------------------|------------|----------------------------------------------------|
| cluster_df        | data.frame | Data frame with sample IDs and cluster assignments |
| similarity_matrix | matrix     | Sample similarity matrix                           |
| normalize_matrix  | logical    | Whether to normalize the matrix (default: FALSE)   |

**Returns:** Silhouette object with per-sample scores.

## 2.11 concordanceNetworkNMI\_ANF

```
concordanceNetworkNMI_ANF(Wall, C, type)
```

**Description:** Calculates concordance between networks and clustering using NMI for ANF method.

**Parameters:**

| Parameter | Type      | Description                     |
|-----------|-----------|---------------------------------|
| Wall      | list      | List of affinity matrices       |
| C         | integer   | Number of clusters              |
| type      | character | Type of concordance calculation |

**Returns:** NMI-based concordance scores.

## 2.12 create\_annot\_barchart

```
create_annot_barchart(plotdata = NULL, fill = NULL, title = NULL,  
  xlab = NULL, ylab = NULL, legend.position = "right")
```

**Description:** Creates annotated bar charts for visualizing cluster distributions.

**Parameters:**

| Parameter                    | Type                    | Description                        |
|------------------------------|-------------------------|------------------------------------|
| <code>plotdata</code>        | <code>data.frame</code> | Data for plotting                  |
| <code>fill</code>            | <code>character</code>  | Column name for fill color         |
| <code>title</code>           | <code>character</code>  | Plot title                         |
| <code>xlab</code>            | <code>character</code>  | X-axis label                       |
| <code>ylab</code>            | <code>character</code>  | Y-axis label                       |
| <code>legend.position</code> | <code>character</code>  | Legend position (default: “right”) |

**Returns:** ggplot2 object.

---

### 2.13 `create__density__curve`

```
create_density_curve(matrix)
```

**Description:** Creates a density curve plot for matrix values.

**Parameters:**

| Parameter           | Type                | Description       |
|---------------------|---------------------|-------------------|
| <code>matrix</code> | <code>matrix</code> | Input data matrix |

**Returns:** ggplot2 density plot object.

---

### 2.14 `create__density__plot__color`

```
create_density_plot_color(matrix)
```

**Description:** Creates a colored density plot for matrix value distributions.

**Parameters:**

| Parameter           | Type                | Description       |
|---------------------|---------------------|-------------------|
| <code>matrix</code> | <code>matrix</code> | Input data matrix |

**Returns:** ggplot2 density plot with color gradient.

---

### 2.15 `create__MO__heatmap`

```
create_MO_heatmap(matrix = NULL, algorithm = NULL, clust_res = NULL,
                  data_name = NULL, fig.path = getwd(), ...)
```

**Description:** Creates multi-omics heatmaps with cluster annotations.

**Parameters:**

| Parameter | Type       | Description                                |
|-----------|------------|--------------------------------------------|
| matrix    | matrix     | Data or similarity matrix to plot          |
| algorithm | character  | Algorithm name for labeling                |
| clust_res | data.frame | Clustering results with sample assignments |
| data_name | character  | Name for the data type                     |
| fig.path  | character  | Output path for figures                    |
| ...       |            | Additional ComplexHeatmap parameters       |

**Returns:** ComplexHeatmap object saved to file.

## 2.16 dist2.cimlr\_mod2

```
dist2.cimlr_mod2(x, c = NA, name = NULL, method = "sqeuclidean")
```

**Description:** Modified squared Euclidean distance calculation for CIMLR.

**Parameters:**

| Parameter | Type      | Description                              |
|-----------|-----------|------------------------------------------|
| x         | matrix    | Data matrix                              |
| c         | matrix    | Centers (optional)                       |
| name      | character | Data name for labeling                   |
| method    | character | Distance method (default: "sqeuclidean") |

**Returns:** Distance matrix.

## 2.17 dist2.cimlr.weight\_mod

```
dist2.cimlr.weight_mod(x, weight, method = "sqeuclidean")
```

**Description:** Weighted distance calculation for CIMLR.

**Parameters:**

| Parameter           | Type      | Description     |
|---------------------|-----------|-----------------|
| <code>x</code>      | matrix    | Data matrix     |
| <code>weight</code> | numeric   | Feature weights |
| <code>method</code> | character | Distance method |

**Returns:** Weighted distance matrix.

## 2.18 `divide_by_quantile_normalize`

```
divide_by_quantile_normalize(matrix, norm_quant = 0.05)
```

**Description:** Normalizes matrix by dividing by a quantile value.

**Parameters:**

| Parameter               | Type    | Description                                |
|-------------------------|---------|--------------------------------------------|
| <code>matrix</code>     | matrix  | Input matrix                               |
| <code>norm_quant</code> | numeric | Quantile for normalization (default: 0.05) |

**Returns:** Normalized matrix.

## 2.19 `estimateNumberOfClustersGivenGraph_mod`

```
estimateNumberOfClustersGivenGraph_mod(W, NUMC = 2:5)
```

**Description:** Estimates optimal cluster number using eigengap and rotation cost heuristics.

**Parameters:**

| Parameter         | Type    | Description                                         |
|-------------------|---------|-----------------------------------------------------|
| <code>W</code>    | matrix  | Affinity/similarity matrix                          |
| <code>NUMC</code> | integer | Range of cluster numbers to evaluate (default: 2:5) |

**Returns:** List with optimal K estimates.

## 2.20 `fetch_citation`

```
fetch_citation(algorithm)
```

**Description:** Retrieves citation information for a given algorithm.

**Parameters:**

| Parameter              | Type      | Description    |
|------------------------|-----------|----------------|
| <code>algorithm</code> | character | Algorithm name |

**Returns:** Character string with citation.

---

## 2.21 fetch\_in\_a\_nutshell

```
fetch_in_a_nutshell(algorithm)
```

**Description:** Retrieves brief description for a given algorithm.

**Parameters:**

| Parameter              | Type      | Description    |
|------------------------|-----------|----------------|
| <code>algorithm</code> | character | Algorithm name |

**Returns:** Character string with algorithm summary.

---

## 2.22 frobenius\_norm

```
frobenius_norm(mat1, mat2)
```

**Description:** Calculates the Frobenius norm distance between two matrices.

**Parameters:**

| Parameter         | Type   | Description   |
|-------------------|--------|---------------|
| <code>mat1</code> | matrix | First matrix  |
| <code>mat2</code> | matrix | Second matrix |

**Returns:** Numeric Frobenius norm value.

---

## 2.23 getMoHeatmap\_prenorm\_quart

```
getMoHeatmap_prenorm_quant(data = NULL, is.binary = c(FALSE, FALSE, FALSE, FALSE, FALSE),
                           clust.res = NULL, ...)
```

**Description:** Generates multi-omics heatmap with quartile-based pre-normalization.

**Parameters:**

| Parameter              | Type       | Description                         |
|------------------------|------------|-------------------------------------|
| <code>data</code>      | list       | List of omics data matrices         |
| <code>is.binary</code> | logical    | Vector indicating binary data types |
| <code>clust.res</code> | data.frame | Clustering results                  |
| <code>...</code>       |            | Additional heatmap parameters       |

**Returns:** ComplexHeatmap object.

## 2.24 linear\_quantile\_normalize

```
linear_quantile_normalize(matrix, norm_quant = 0.05)
```

**Description:** Linear quantile normalization scaling matrix to [0,1] based on quantiles.

**Parameters:**

| Parameter               | Type    | Description                        |
|-------------------------|---------|------------------------------------|
| <code>matrix</code>     | matrix  | Input matrix                       |
| <code>norm_quant</code> | numeric | Quantile threshold (default: 0.05) |

**Returns:** Normalized matrix.

## 2.25 mds\_from\_original\_matrix

```
mds_from_original_matrix(matrix = NULL, dist_method = NULL, algorithm = NULL,
                          clust_res = NULL, fig.path = getwd(), ...)
```

**Description:** Performs MDS from original data matrix for visualization.

**Parameters:**

| Parameter                | Type      | Description             |
|--------------------------|-----------|-------------------------|
| <code>matrix</code>      | matrix    | Input data matrix       |
| <code>dist_method</code> | character | Distance method for MDS |

| Parameter              | Type       | Description                 |
|------------------------|------------|-----------------------------|
| <code>algorithm</code> | character  | Algorithm name for labeling |
| <code>clust_res</code> | data.frame | Clustering results          |
| <code>fig.path</code>  | character  | Output path                 |

**Returns:** MDS plot saved to file.

## 2.26 MOVICS\_\_jaccard\_index

```
MOVICS__jaccard_index(clust1, clust2)
```

**Description:** Calculates Jaccard index between two cluster assignments.

**Parameters:**

| Parameter           | Type   | Description               |
|---------------------|--------|---------------------------|
| <code>clust1</code> | vector | First cluster assignment  |
| <code>clust2</code> | vector | Second cluster assignment |

**Returns:** Numeric Jaccard index (0 to 1).

## 2.27 multiple.kernel.cimlr\_mod

```
multiple.kernel.cimlr_mod(x, cores.ratio = 1, name = NULL, method = "sqeuclidean")
```

**Description:** Generates multiple kernels for CIMLR integration.

**Parameters:**

| Parameter                | Type      | Description                            |
|--------------------------|-----------|----------------------------------------|
| <code>x</code>           | matrix    | Data matrix                            |
| <code>cores.ratio</code> | numeric   | Ratio of cores for parallel processing |
| <code>name</code>        | character | Data name                              |
| <code>method</code>      | character | Distance method                        |

**Returns:** List of kernel matrices.

## 2.28 multiple.kernel.cimlr.weight\_mod



```
multiple.kernel.cimlr.weight_mod(x, cores.ratio = 0, weight, method = "sqeuclidean")
```

**Description:** Weighted multiple kernel generation for CIMLR.

**Parameters:**

| Parameter                | Type      | Description     |
|--------------------------|-----------|-----------------|
| <code>x</code>           | matrix    | Data matrix     |
| <code>weight</code>      | numeric   | Feature weights |
| <code>cores.ratio</code> | numeric   | Ratio of cores  |
| <code>method</code>      | character | Distance method |

**Returns:** Weighted kernel list.

---

## 2.29 nemo.affinity.graph\_mod

```
nemo.affinity.graph_mod(raw.data, k = NA, sigma = 0.5,  
  binary_flags = rep("No", length(raw.data)),  
  binary_distance = NULL)
```

**Description:** Modified NEMO affinity graph construction supporting mixed data types.

**Parameters:**

| Parameter                    | Type      | Description                   |
|------------------------------|-----------|-------------------------------|
| <code>raw.data</code>        | list      | List of data matrices         |
| <code>k</code>               | integer   | Number of neighbors           |
| <code>sigma</code>           | numeric   | Kernel width (default: 0.5)   |
| <code>binary_flags</code>    | character | Vector indicating binary data |
| <code>binary_distance</code> | character | Distance for binary data      |

**Returns:** Affinity matrix.

---

## 2.30 normalize\_affinity\_matrix

```
normalize_affinity_matrix(W)
```

**Description:** Normalizes an affinity matrix to have row sums of 1.

**Parameters:**

| Parameter | Type   | Description     |
|-----------|--------|-----------------|
| <i>W</i>  | matrix | Affinity matrix |

**Returns:** Row-normalized affinity matrix.

## 2.31 `pca_from_original_matrix`

```
pca_from_original_matrix(mydata = NULL, algorithm = NULL, clust_res = NULL,
                        data_name = NULL, fig.path = getwd(), ...)
```

**Description:** Performs PCA on original data and creates visualization with cluster colors.

**Parameters:**

| Parameter        | Type       | Description                 |
|------------------|------------|-----------------------------|
| <i>mydata</i>    | matrix     | Input data matrix           |
| <i>algorithm</i> | character  | Algorithm name for labeling |
| <i>clust_res</i> | data.frame | Clustering results          |
| <i>data_name</i> | character  | Data type name              |
| <i>fig.path</i>  | character  | Output path                 |

**Returns:** PCA plot saved to file.

## 2.32 `pca_from_sim_matrix`

```
pca_from_sim_matrix(sim_matrix = NULL, algorithm = NULL, clust_res = NULL,
                    data_name = NULL, fig.path = getwd(), ...)
```

**Description:** Performs PCA-based visualization from a similarity matrix.

**Parameters:**

| Parameter         | Type       | Description                |
|-------------------|------------|----------------------------|
| <i>sim_matrix</i> | matrix     | Similarity/affinity matrix |
| <i>algorithm</i>  | character  | Algorithm name             |
| <i>clust_res</i>  | data.frame | Clustering results         |
| <i>data_name</i>  | character  | Data type name             |
| <i>fig.path</i>   | character  | Output path                |

**Returns:** PCA plot from similarity matrix.

## 2.33 pearson\_correlation

```
pearson_correlation(mat1, mat2)
```

**Description:** Calculates Pearson correlation between vectorized matrices.

**Parameters:**

| Parameter | Type   | Description   |
|-----------|--------|---------------|
| mat1      | matrix | First matrix  |
| mat2      | matrix | Second matrix |

**Returns:** Pearson correlation coefficient.

---

## 2.34 plot\_MOFA\_cumul\_var\_exp

```
plot_MOFA_cumul_var_exp(object, factor_range = NULL, ...)
```

**Description:** Plots cumulative variance explained by MOFA factors.

**Parameters:**

| Parameter    | Type        | Description                  |
|--------------|-------------|------------------------------|
| object       | MOFA object | Trained MOFA model           |
| factor_range | integer     | Range of factors to plot     |
| ...          |             | Additional ggplot parameters |

**Returns:** ggplot2 cumulative variance plot.

---

## 2.35 plot\_MOFA\_var\_exp

```
plot_MOFA_var_exp(object, factor_range = NULL, ...)
```

**Description:** Plots variance explained per view and factor for MOFA models.

**Parameters:**

| Parameter    | Type        | Description                  |
|--------------|-------------|------------------------------|
| object       | MOFA object | Trained MOFA model           |
| factor_range | integer     | Range of factors to plot     |
| ...          |             | Additional ggplot parameters |

**Returns:** ggplot2 variance explained heatmap.

---

## 2.36 rankFeaturesByNMI\_parallelly

```
rankFeaturesByNMI_parallelly(data, W, ncores = detectCores() - 1,  
                             binary = FALSE, nn = NULL, sigma)
```

**Description:** Ranks features by NMI contribution using parallel processing.

**Parameters:**

| Parameter | Type    | Description                                  |
|-----------|---------|----------------------------------------------|
| data      | list    | Single-element list with data matrix         |
| W         | matrix  | Similarity matrix                            |
| ncores    | integer | Number of cores (default: detectCores() - 1) |
| binary    | logical | Binary data flag                             |
| nn        | integer | Number of neighbors                          |
| sigma     | numeric | Kernel width                                 |

**Returns:** Data frame with feature NMI scores.

## 2.37 rankFeaturesByNMI\_parallelly\_ANF

```
rankFeaturesByNMI_parallelly_ANF(data, W, ncores = detectCores() - 1,  
                                 binary = FALSE, ...)
```

**Description:** ANF-specific feature ranking by NMI.

**Parameters:**

| Parameter | Type    | Description           |
|-----------|---------|-----------------------|
| data      | list    | Data list             |
| W         | matrix  | ANF similarity matrix |
| ncores    | integer | Number of cores       |
| binary    | logical | Binary data flag      |

**Returns:** Feature ranking data frame.

## 2.38 reshape\_Pearson\_for\_tests

```
reshape_Pearson_for_tests(similarities)
```

**Description:** Reshapes Pearson correlation matrix for statistical testing.

**Parameters:**

| Parameter                 | Type   | Description       |
|---------------------------|--------|-------------------|
| <code>similarities</code> | matrix | Similarity matrix |

**Returns:** Reshaped data frame for testing.

---

## 2.39 `standardize_rows`

```
standardize_rows(mat)
```

**Description:** Z-score standardizes each row of a matrix.

**Parameters:**

| Parameter        | Type   | Description  |
|------------------|--------|--------------|
| <code>mat</code> | matrix | Input matrix |

**Returns:** Row-standardized matrix.

---

## 2.40 `TCGAanalyze_survival_custom3`

```
TCGAanalyze_survival_custom3(clinical_data, cluster_col, time_col,  
                               event_col, algorithm = NULL, ...)
```

**Description:** Custom survival analysis for TCGA data with multi-group comparisons.

**Parameters:**

| Parameter                  | Type       | Description                             |
|----------------------------|------------|-----------------------------------------|
| <code>clinical_data</code> | data.frame | Clinical data with survival information |
| <code>cluster_col</code>   | character  | Column with cluster assignments         |
| <code>time_col</code>      | character  | Column with survival time               |
| <code>event_col</code>     | character  | Column with event status                |
| <code>algorithm</code>     | character  | Algorithm name for labeling             |

**Returns:** Survival analysis results with Kaplan-Meier plots.

---

## 3 fast\_pathfindR\_hclust.R

Optimized functions for pathway enrichment term clustering.

### 3.1 cluster\_enriched\_terms\_fast

```
cluster_enriched_terms_fast(enrichment_res, method = "hierarchical",  
                             plot_clusters_graph = TRUE, ...)
```

**Description:** Main function for clustering enriched pathway terms using optimized algorithms.

**Parameters:**

| Parameter           | Type       | Description                                  |
|---------------------|------------|----------------------------------------------|
| enrichment_res      | data.frame | Enrichment results from pathfindR            |
| method              | character  | Clustering method: “hierarchical” or “fuzzy” |
| plot_clusters_graph | logical    | Whether to plot cluster graph                |
| ...                 |            | Additional clustering parameters             |

**Returns:** List with clustered terms and visualization.

---

### 3.2 cluster\_graph\_vis\_fast

```
cluster_graph_vis_fast(clu_obj, kappa_mat, enrichment_res,  
                        kappa_threshold = 0.35, ...)
```

**Description:** Creates fast graph visualization of term clusters.

**Parameters:**

| Parameter       | Type       | Description                 |
|-----------------|------------|-----------------------------|
| clu_obj         | list       | Clustering result object    |
| kappa_mat       | matrix     | Kappa statistic matrix      |
| enrichment_res  | data.frame | Original enrichment results |
| kappa_threshold | numeric    | Threshold for edge creation |

**Returns:** igraph visualization object.

---

### 3.3 create\_kappa\_matrix\_fast

```
create_kappa_matrix_fast(enrichment_res, use_description = FALSE,
                        use_active_snw_genes = FALSE)
```

**Description:** Creates kappa statistic matrix for pathway term similarity using optimized algorithm.

**Parameters:**

| Parameter            | Type       | Description                 |
|----------------------|------------|-----------------------------|
| enrichment_res       | data.frame | Enrichment results          |
| use_description      | logical    | Use pathway descriptions    |
| use_active_snw_genes | logical    | Use active subnetwork genes |

**Returns:** Kappa matrix for term similarity.

### 3.4 fuzzy\_term\_clustering\_fast

```
fuzzy_term_clustering_fast(kappa_mat, enrichment_res,
                          kappa_threshold = 0.35, ...)
```

**Description:** Performs fuzzy clustering on pathway terms.

**Parameters:**

| Parameter       | Type       | Description             |
|-----------------|------------|-------------------------|
| kappa_mat       | matrix     | Kappa similarity matrix |
| enrichment_res  | data.frame | Enrichment results      |
| kappa_threshold | numeric    | Similarity threshold    |

**Returns:** Fuzzy clustering assignments.

### 3.5 hierarchical\_term\_clustering\_fast

```
hierarchical_term_clustering_fast(kappa_mat, enrichment_res,
                                 num_clusters = NULL, ...)
```

**Description:** Performs hierarchical clustering on pathway terms.

**Parameters:**

| Parameter | Type   | Description             |
|-----------|--------|-------------------------|
| kappa_mat | matrix | Kappa similarity matrix |

| Parameter                   | Type                    | Description                       |
|-----------------------------|-------------------------|-----------------------------------|
| <code>enrichment_res</code> | <code>data.frame</code> | Enrichment results                |
| <code>num_clusters</code>   | <code>integer</code>    | Number of clusters (auto if NULL) |

**Returns:** Hierarchical clustering result.

## 4 modified\_\_MOVICS\_\_functions.R

Modified MOVICS package functions for multi-omics analysis.

### 4.1 compAgree2

```
compAgree2(moic.res = NULL, subt2comp = NULL, doPlot = TRUE, ...)
```

**Description:** Compares agreement between two clustering results with visualization.

**Parameters:**

| Parameter              | Type                    | Description                  |
|------------------------|-------------------------|------------------------------|
| <code>moic.res</code>  | <code>list</code>       | MOVICS result object         |
| <code>subt2comp</code> | <code>data.frame</code> | Second clustering to compare |
| <code>doPlot</code>    | <code>logical</code>    | Generate comparison plot     |

**Returns:** Agreement statistics and optional Sankey plot.

### 4.2 compAgree\_single\_algorithm

```
compAgree_single_algorithm(algorithm_name = "CS", moic.res = NULL,
                           subt2comp = NULL, ...)
```

**Description:** Single-algorithm version of clustering agreement comparison.

**Parameters:**

| Parameter                   | Type                    | Description          |
|-----------------------------|-------------------------|----------------------|
| <code>algorithm_name</code> | <code>character</code>  | Algorithm identifier |
| <code>moic.res</code>       | <code>list</code>       | Clustering result    |
| <code>subt2comp</code>      | <code>data.frame</code> | Reference clustering |

**Returns:** Agreement metrics.



### 4.3 compClinvar2

```
compClinvar2(moic.res = NULL, var2comp = NULL, strata = NULL,  
             factorVars = NULL, ...)
```

**Description:** Compares clinical variables across subtypes.

**Parameters:**

| Parameter  | Type       | Description             |
|------------|------------|-------------------------|
| moic.res   | list       | MOVICS result           |
| var2comp   | data.frame | Clinical variables      |
| strata     | character  | Stratification variable |
| factorVars | character  | Factor variables        |

**Returns:** Clinical comparison table with p-values.

---

### 4.4 compClinvar\_ordinal\_single\_algorithm

```
compClinvar_ordinal_single_algorithm(algorithm_name = "CS", moic.res = NULL,  
                                     var2comp = NULL, ordinal_vars = NULL, ...)
```

**Description:** Compares ordinal clinical variables using appropriate tests.

**Parameters:**

| Parameter      | Type       | Description            |
|----------------|------------|------------------------|
| algorithm_name | character  | Algorithm identifier   |
| moic.res       | list       | Clustering result      |
| var2comp       | data.frame | Clinical variables     |
| ordinal_vars   | character  | Ordinal variable names |

**Returns:** Ordinal comparison results.

---

### 4.5 compClinvar\_single\_algorithm

```
compClinvar_single_algorithm(algorithm_name = "CS", moic.res = NULL,  
                             var2comp = NULL, ...)
```

**Description:** Single-algorithm clinical variable comparison.

**Parameters:**

| Parameter                   | Type       | Description          |
|-----------------------------|------------|----------------------|
| <code>algorithm_name</code> | character  | Algorithm identifier |
| <code>moic.res</code>       | list       | Clustering result    |
| <code>var2comp</code>       | data.frame | Clinical variables   |

**Returns:** Comparison table.

---

## 4.6 `compDrugsen_single_algorithm`

```
compDrugsen_single_algorithm(algorithm_name = "CS", moic.res = NULL,
                             norm.expr = NULL, drugs = c("Cisplatin", "Paclitaxel"), ...)
```

**Description:** Compares predicted drug sensitivity across subtypes.

**Parameters:**

| Parameter                   | Type      | Description           |
|-----------------------------|-----------|-----------------------|
| <code>algorithm_name</code> | character | Algorithm identifier  |
| <code>moic.res</code>       | list      | Clustering result     |
| <code>norm.expr</code>      | matrix    | Normalized expression |
| <code>drugs</code>          | character | Drug names to analyze |

**Returns:** Drug sensitivity comparison.

---

## 4.7 `compFGA_mod`

```
compFGA_mod(moic.res = NULL, segment = NULL, iscopynumber = FALSE,
             ga_column = NULL, ...)
```

**Description:** Compares Fraction of Genome Altered (FGA) across subtypes.

**Parameters:**

| Parameter                 | Type       | Description                 |
|---------------------------|------------|-----------------------------|
| <code>moic.res</code>     | list       | Clustering result           |
| <code>segment</code>      | data.frame | Segmentation data           |
| <code>iscopynumber</code> | logical    | Whether data is copy number |
| <code>ga_column</code>    | character  | Genome alteration column    |

**Returns:** FGA comparison with visualization.

---

## 4.8 compFGA\_optimized

```
compFGA_optimized(moic.res = NULL, segment = NULL, iscopynumber = FALSE,  
                  ga_column = NULL, ...)
```

**Description:** Optimized FGA comparison with improved performance.

**Parameters:**

| Parameter             | Type | Description |
|-----------------------|------|-------------|
| (Same as compFGA_mod) |      |             |

**Returns:** FGA comparison results.

---

## 4.9 compMut\_single\_algorithm

```
compMut_single_algorithm(algorithm_name = "CS", moic.res = NULL,  
                         mut.matrix = NULL, freq.cutoff = 0.05, ...)
```

**Description:** Compares mutation frequencies across subtypes.

**Parameters:**

| Parameter      | Type      | Description                      |
|----------------|-----------|----------------------------------|
| algorithm_name | character | Algorithm identifier             |
| moic.res       | list      | Clustering result                |
| mut.matrix     | matrix    | Mutation matrix                  |
| freq.cutoff    | numeric   | Frequency cutoff (default: 0.05) |

**Returns:** Mutation comparison with oncoprint.

---

## 4.10 compMut\_single\_algorithm\_sc

```
compMut_single_algorithm_sc(algorithm_name = "CS", moic.res = NULL,  
                           mut.matrix = NULL, ...)
```

**Description:** Single-cell adapted mutation comparison.

**Parameters:**

| Parameter                          | Type | Description |
|------------------------------------|------|-------------|
| (Same as compMut_single_algorithm) |      |             |

**Returns:** Mutation comparison for single-cell data.

#### 4.11 compSurv\_ext

```
compSurv_ext(moic.res = NULL, surv.info = NULL, convt.time = "d", ...)
```

**Description:** Extended survival comparison with multiple tests.

**Parameters:**

| Parameter  | Type       | Description                     |
|------------|------------|---------------------------------|
| moic.res   | list       | Clustering result               |
| surv.info  | data.frame | Survival information            |
| convt.time | character  | Time conversion ("d", "m", "y") |

**Returns:** Comprehensive survival analysis.

#### 4.12 getMoHeatmap\_single\_algorithm

```
getMoHeatmap_single_algorithm(algorithm_name = "CS", data = NULL,
                              is.binary = c(FALSE), ...)
```

**Description:** Generates multi-omics heatmap for single algorithm result.

**Parameters:**

| Parameter      | Type      | Description          |
|----------------|-----------|----------------------|
| algorithm_name | character | Algorithm identifier |
| data           | list      | Multi-omics data     |
| is.binary      | logical   | Binary data flags    |

**Returns:** ComplexHeatmap object.

#### 4.13 getMoHeatmap\_single\_algorithm2

```
getMoHeatmap_single_algorithm2(algorithm_name = "CS", data = NULL, ...)
```

**Description:** Alternative multi-omics heatmap generation.

**Parameters:**

| Parameter                               | Type | Description |
|-----------------------------------------|------|-------------|
| (Same as getMoHeatmap_single_algorithm) |      |             |

**Returns:** Heatmap visualization.

---

## 4.14 getSilhouette\_ggplot

```
getSilhouette_ggplot(sil = NULL, clust.col = NULL, fig.path = getwd(),  
                      fig.name = "silhouette_ggplot", algorithm = "", ...)
```

**Description:** Creates ggplot2-based silhouette plot.

**Parameters:**

| Parameter | Type       | Description               |
|-----------|------------|---------------------------|
| sil       | silhouette | Silhouette object         |
| clust.col | character  | Cluster colors            |
| fig.path  | character  | Output path               |
| fig.name  | character  | Output filename           |
| algorithm | character  | Algorithm name for labels |

**Returns:** ggplot2 silhouette visualization.

---

## 4.15 ntp\_mod

```
ntp_mod(emat, templates, nPerm = 1000, distance = "cosine", ...)
```

**Description:** Modified Nearest Template Prediction algorithm.

**Parameters:**

| Parameter | Type    | Description            |
|-----------|---------|------------------------|
| emat      | matrix  | Expression matrix      |
| templates | list    | Template signatures    |
| nPerm     | integer | Number of permutations |

| Parameter             | Type      | Description     |
|-----------------------|-----------|-----------------|
| <code>distance</code> | character | Distance metric |

**Returns:** NTP predictions with p-values.

## 4.16 `plot_pathway_heatmaps`

```
plot_pathway_heatmaps(gsea.lists, norm.expr = NULL, algorithm_name = NULL, ...)
```

**Description:** Creates pathway-level heatmaps from GSEA results.

**Parameters:**

| Parameter                   | Type      | Description           |
|-----------------------------|-----------|-----------------------|
| <code>gsea.lists</code>     | list      | GSEA result lists     |
| <code>norm.expr</code>      | matrix    | Normalized expression |
| <code>algorithm_name</code> | character | Algorithm identifier  |

**Returns:** Pathway heatmap visualizations.

## 4.17 `prepare_gsea_output_for_hclust`

```
prepare_gsea_output_for_hclust(gsea_output, dgea_output_name_style = "", ...)
```

**Description:** Prepares GSEA output for hierarchical clustering analysis.

**Parameters:**

| Parameter                           | Type      | Description            |
|-------------------------------------|-----------|------------------------|
| <code>gsea_output</code>            | list      | GSEA results           |
| <code>dgea_output_name_style</code> | character | DGEA naming convention |

**Returns:** Formatted data for pathway clustering.

## 4.18 `runDEA_mod`

```
runDEA_mod(dea.method = c("deseq2", "edger", "limma"), expr = NULL,
           moic.res = NULL, ...)
```

**Description:** Modified differential expression analysis wrapper.

**Parameters:**

| Parameter  | Type      | Description       |
|------------|-----------|-------------------|
| dea.method | character | DEA method to use |
| expr       | matrix    | Expression matrix |
| moic.res   | list      | Clustering result |

**Returns:** DEA results for all pairwise comparisons.

#### 4.19 runGSEA\_mod

```
runGSEA_mod(moic.res = NULL, dea.method = c("deseq2", "edger", "limma"), ...)
```

**Description:** Modified GSEA analysis for MOVICS results.

**Parameters:**

| Parameter  | Type      | Description       |
|------------|-----------|-------------------|
| moic.res   | list      | Clustering result |
| dea.method | character | DEA method used   |

**Returns:** GSEA enrichment results.

#### 4.20 runGSEA\_mod\_4.4

```
runGSEA_mod_4.4(moic.res = NULL, dea.method = c("deseq2", "edger", "limma"), ...)
```

**Description:** GSEA modified for R 4.4 compatibility.

**Parameters:**

| Parameter             | Type | Description |
|-----------------------|------|-------------|
| (Same as runGSEA_mod) |      |             |

**Returns:** R 4.4 compatible GSEA results.

## 4.21 runGSEA\_mod\_4.4\_single\_algorithm

```
runGSEA_mod_4.4_single_algorithm(algorithm_name = "CS", moic.res = NULL, ...)
```

**Description:** Single-algorithm GSEA for R 4.4.

**Parameters:**

| Parameter      | Type      | Description          |
|----------------|-----------|----------------------|
| algorithm_name | character | Algorithm identifier |
| moic.res       | list      | Clustering result    |

**Returns:** GSEA results with algorithm prefix.

---

## 4.22 runGSVA\_mod\_4.4

```
runGSVA_mod_4.4(moic.res = NULL, norm.expr = NULL, gset.gmt.path = NULL, ...)
```

**Description:** Modified GSVA analysis for R 4.4.

**Parameters:**

| Parameter     | Type      | Description           |
|---------------|-----------|-----------------------|
| moic.res      | list      | Clustering result     |
| norm.expr     | matrix    | Normalized expression |
| gset.gmt.path | character | Path to GMT file      |

**Returns:** GSVA enrichment scores.

---

## 4.23 runGSVA\_mod\_4.4\_single\_algorithm

```
runGSVA_mod_4.4_single_algorithm(algorithm_name = "CS", moic.res = NULL, ...)
```

**Description:** Single-algorithm GSVA for R 4.4.

**Parameters:**

| Parameter                                     | Type | Description |
|-----------------------------------------------|------|-------------|
| (Same as runGSVA_mod_4.4 with algorithm_name) |      |             |

**Returns:** GSVA results with algorithm prefix.

---



## 4.24 runKappa\_single\_algorithm

```
runKappa_single_algorithm(algorithm_name = "CS", moic.res = NULL, ...)
```

**Description:** Calculates Cohen's Kappa for clustering agreement.

**Parameters:**

| Parameter      | Type      | Description          |
|----------------|-----------|----------------------|
| algorithm_name | character | Algorithm identifier |
| moic.res       | list      | Clustering result    |

**Returns:** Kappa statistics.

---

## 4.25 runMarker\_mod\_4.4

```
runMarker_mod_4.4(moic.res = NULL, dea.method = c("deseq2", "edger", "limma"), ...)
```

**Description:** Marker gene identification for R 4.4.

**Parameters:**

| Parameter  | Type      | Description       |
|------------|-----------|-------------------|
| moic.res   | list      | Clustering result |
| dea.method | character | DEA method        |

**Returns:** Marker genes per subtype.

---

## 4.26 runMarker\_single\_algorithm

```
runMarker_single_algorithm(algorithm_name = "CS", moic.res = NULL,  
                           dea.method = c("deseq2", "edger", "limma"), ...)
```

**Description:** Single-algorithm marker identification.

**Parameters:**

| Parameter      | Type      | Description          |
|----------------|-----------|----------------------|
| algorithm_name | character | Algorithm identifier |
| moic.res       | list      | Clustering result    |
| dea.method     | character | DEA method           |

**Returns:** Marker genes with algorithm prefix.

---

## 4.27 runMarker\_single\_algorithm\_no\_export

```
runMarker_single_algorithm_no_export(algorithm_name = "CS", moic.res = NULL, ...)
```

**Description:** Marker identification without file export.

**Parameters:**

| Parameter                            | Type | Description |
|--------------------------------------|------|-------------|
| (Same as runMarker_single_algorithm) |      |             |

**Returns:** In-memory marker results.

---

## 4.28 runNTP\_mod

```
runNTP_mod(expr = NULL, templates = NULL, scaleFlag = TRUE,  
            centerFlag = TRUE, ...)
```

**Description:** Modified Nearest Template Prediction.

**Parameters:**

| Parameter  | Type    | Description        |
|------------|---------|--------------------|
| expr       | matrix  | Expression matrix  |
| templates  | list    | Template gene sets |
| scaleFlag  | logical | Scale data         |
| centerFlag | logical | Center data        |

**Returns:** NTP classification results.

---

## 4.29 runPAM\_single\_algorithm

```
runPAM_single_algorithm(algorithm_name = "CS", moic.res = NULL,  
                         norm.expr = NULL, ...)
```

**Description:** PAM (Prediction Analysis for Microarrays) for subtype prediction.

**Parameters:**

| Parameter      | Type      | Description          |
|----------------|-----------|----------------------|
| algorithm_name | character | Algorithm identifier |

| Parameter              | Type   | Description           |
|------------------------|--------|-----------------------|
| <code>moic.res</code>  | list   | Clustering result     |
| <code>norm.expr</code> | matrix | Normalized expression |

**Returns:** PAM classifier and predictions.

#### 4.30 `subHeatmap_mod`

```
subHeatmap_mod(emat, res, templates, keepN = TRUE, labRow = NULL, ...)
```

**Description:** Modified subtype heatmap visualization.

**Parameters:**

| Parameter              | Type      | Description         |
|------------------------|-----------|---------------------|
| <code>emat</code>      | matrix    | Expression matrix   |
| <code>res</code>       | list      | NTP results         |
| <code>templates</code> | list      | Template signatures |
| <code>keepN</code>     | logical   | Keep all samples    |
| <code>labRow</code>    | character | Row labels          |

**Returns:** Subtype heatmap.

#### 4.31 `twoclassdeseq2_mod`

```
twoclassdeseq2_mod(moic.res = NULL, countsTable = NULL, prefix = NULL, ...)
```

**Description:** Two-class DESeq2 differential expression.

**Parameters:**

| Parameter                | Type      | Description       |
|--------------------------|-----------|-------------------|
| <code>moic.res</code>    | list      | Clustering result |
| <code>countsTable</code> | matrix    | Count matrix      |
| <code>prefix</code>      | character | Output prefix     |

**Returns:** DESeq2 results.

#### 4.32 `twoclassedger_mod`

```
twoclassedgeR_mod(moic.res = NULL, countsTable = NULL, prefix = NULL, ...)
```

**Description:** Two-class edgeR differential expression.

**Parameters:**

| Parameter                    | Type | Description |
|------------------------------|------|-------------|
| (Same as twoclassdeseq2_mod) |      |             |

**Returns:** edgeR results.

---

### 4.33 twoclasslimma\_mod

```
twoclasslimma_mod(moic.res = NULL, norm.expr = NULL, prefix = NULL, ...)
```

**Description:** Two-class limma differential expression.

**Parameters:**

| Parameter | Type      | Description           |
|-----------|-----------|-----------------------|
| moic.res  | list      | Clustering result     |
| norm.expr | matrix    | Normalized expression |
| prefix    | character | Output prefix         |

**Returns:** limma results.

---

## 5 Spectrum\_functions.R

Modified Spectrum spectral clustering functions.

### 5.1 CNN\_kernel\_mod

```
CNN_kernel_mod(mat, NN = 3, NN2 = 7, distance = "euclidean")
```

**Description:** Modified Continuous Nearest Neighbor kernel construction.

**Parameters:**

| Parameter             | Type      | Description                              |
|-----------------------|-----------|------------------------------------------|
| <code>mat</code>      | matrix    | Data matrix                              |
| <code>NN</code>       | integer   | Number of nearest neighbors (default: 3) |
| <code>NN2</code>      | integer   | Extended neighbors (default: 7)          |
| <code>distance</code> | character | Distance metric                          |

**Returns:** CNN kernel matrix.

## 5.2 kernfinder\_local\_mod

```
kernfinder_local_mod(data, maxk = 10, fontsize = 12, silent = FALSE, ...)
```

**Description:** Local kernel parameter finder for Spectrum.

**Parameters:**

| Parameter             | Type    | Description                     |
|-----------------------|---------|---------------------------------|
| <code>data</code>     | matrix  | Data matrix                     |
| <code>maxk</code>     | integer | Maximum k to test (default: 10) |
| <code>fontsize</code> | integer | Plot font size                  |
| <code>silent</code>   | logical | Suppress output                 |

**Returns:** Optimal local kernel parameters.

## 5.3 kernfinder\_mine\_mod

```
kernfinder_mine_mod(data, maxk = 10, fontsize = 12, silent = FALSE, ...)
```

**Description:** Mining-based kernel parameter finder.

**Parameters:**

| Parameter                      | Type | Description |
|--------------------------------|------|-------------|
| (Same as kernfinder_local_mod) |      |             |

**Returns:** Optimal kernel parameters.

## 5.4 rbfkernel\_b\_mod

```
rbfkernel_b_mod(mat, K = 3, sigma = 1, distance = "euclidean")
```

**Description:** Modified RBF kernel construction.

**Parameters:**

| Parameter             | Type      | Description                      |
|-----------------------|-----------|----------------------------------|
| <code>mat</code>      | matrix    | Data matrix                      |
| <code>K</code>        | integer   | Number of neighbors (default: 3) |
| <code>sigma</code>    | numeric   | Kernel width (default: 1)        |
| <code>distance</code> | character | Distance metric                  |

**Returns:** RBF kernel matrix.

---

## 5.5 Spectrum\_bin\_and\_par

```
Spectrum_bin_and_par(data, method = "CNN", maxk = 10, ...)
```

**Description:** Main Spectrum clustering with binary data support and parallel processing.

**Parameters:**

| Parameter           | Type      | Description                    |
|---------------------|-----------|--------------------------------|
| <code>data</code>   | list      | List of data matrices          |
| <code>method</code> | character | Kernel method ("CNN", "RBF")   |
| <code>maxk</code>   | integer   | Maximum clusters to test       |
| <code>...</code>    |           | Additional Spectrum parameters |

**Returns:** Spectrum clustering results with cluster assignments and metrics.

---

## 6 single\_algorithm/ Helper Functions

Custom helper functions defined within individual algorithm scripts in `Scripts/single_algorithm/`. These functions are documented in `Scripts/single_algorithm/single_algorithm_additional_docs.R`.

### 6.1 MAD-Based Weighting Functions

#### 6.1.1 mad\_based\_weights

```
mad_based_weights(X)
```

**Description:** Calculates Median Absolute Deviation (MAD) based weights for features in a data matrix. Used to assign higher importance to features with greater variability across samples.

**Parameters:**

| Parameter | Type   | Description                                                 |
|-----------|--------|-------------------------------------------------------------|
| X         | matrix | Numeric matrix with features in rows and samples in columns |

**Returns:** Named numeric vector of normalized weights (sum = 1).

**Source:** `ab-SNF.R` (line ~108), `wMKL.R` (line ~160)

---

#### 6.1.2 weighted\_euclidean\_dist

```
weighted_euclidean_dist(X, w)
```

**Description:** Calculates a weighted Euclidean distance matrix between samples, where each feature contribution is scaled by the square root of its weight.

**Parameters:**

| Parameter | Type    | Description                                                 |
|-----------|---------|-------------------------------------------------------------|
| X         | matrix  | Numeric matrix with features in rows and samples in columns |
| w         | numeric | Named vector of feature weights                             |

**Returns:** A square distance matrix of dimension  $\text{ncol}(X) \times \text{ncol}(X)$ .

**Source:** `ab-SNF.R` (line ~120)

---

#### 6.1.3 weighted\_hamming\_fraction

```
weighted_hamming_fraction(binary_mat, features, drivers, driver_weight, nondriver_weight)
```

**Description:** Calculates a weighted Hamming distance matrix for binary data, assigning different weights to driver genes vs. non-driver genes.

**Parameters:**

| Parameter                     | Type      | Description                               |
|-------------------------------|-----------|-------------------------------------------|
| <code>binary_mat</code>       | matrix    | Binary (0/1) matrix with features in rows |
| <code>features</code>         | character | Vector of feature names                   |
| <code>drivers</code>          | character | Vector of driver gene identifiers         |
| <code>driver_weight</code>    | numeric   | Weight for driver gene mismatches         |
| <code>nondriver_weight</code> | numeric   | Weight for non-driver gene mismatches     |

**Returns:** A square distance matrix.

**Source:** `ab-SNF.R` (line ~133)

---

#### 6.1.4 weighted\_mutations

```
weighted_mutations(features, drivers, driver_weight, nondriver_weight)
```

**Description:** Assigns weights to features based on whether they are known driver genes, normalizing so the total weight sums to 1.

**Parameters:**

| Parameter                     | Type      | Description                             |
|-------------------------------|-----------|-----------------------------------------|
| <code>features</code>         | character | Vector of all feature names             |
| <code>drivers</code>          | character | Vector of known driver gene identifiers |
| <code>driver_weight</code>    | numeric   | Weight for driver genes                 |
| <code>nondriver_weight</code> | numeric   | Weight for non-driver genes             |

**Returns:** Numeric vector of normalized weights (sum = 1).

**Source:** `wMKL.R` (line ~176)

---

## 6.2 Matrix Similarity Functions

### 6.2.1 compute\_matrix\_similarity

```
compute_matrix_similarity(matrices)
```



**Description:** Calculates both Frobenius norm distance and Pearson correlation between all pairs of matrices in a list. Used to assess similarity between affinity matrices computed under different hyperparameter settings.

**Parameters:**

| Parameter             | Type | Description                                            |
|-----------------------|------|--------------------------------------------------------|
| <code>matrices</code> | list | Named list of numeric matrices of identical dimensions |

**Returns:** List with two elements: `Frobenius` (distance matrix) and `Pearson` (correlation matrix).

**Source:** `ab-SNF.R`, `ANF.R`, `RWR-F.R`, `RWR-NF.R`, `SNF.R`, `Spectrum.R`

## 6.3 Spectral Clustering Functions

### 6.3.1 `spectralClustering_eig`

```
spectralClustering_eig(affinity, K, type = 3)
```

**Description:** Performs spectral clustering on an affinity matrix and returns both cluster assignments and the eigenvector embedding. Modified version of `SNFtool::spectralClustering`.

**Parameters:**

| Parameter             | Type    | Description                                                                   |
|-----------------------|---------|-------------------------------------------------------------------------------|
| <code>affinity</code> | matrix  | Square symmetric affinity matrix ( $n \times n$ )                             |
| <code>K</code>        | integer | Number of clusters                                                            |
| <code>type</code>     | integer | Laplacian normalization: 1=unnormalized, 2=random walk, 3=symmetric (default) |

**Returns:** Data frame with eigenvector columns, `Cluster` assignments, and `Sample.ID`.

**Source:** `ab-SNF.R` (line ~758), `SNF.R` (line ~693), `NEMO.R` (line ~575)

### 6.3.2 `spectral_clustering_eig`

```
spectral_clustering_eig(affinity, type = "rw", optk)
```

**Description:** ANF-style spectral clustering using random-walk normalized Laplacian with eigenvector extraction.

**Parameters:**

| Parameter       | Type      | Description                               |
|-----------------|-----------|-------------------------------------------|
| <b>affinity</b> | matrix    | Square symmetric affinity matrix          |
| <b>type</b>     | character | Normalization type (“rw” for random walk) |
| <b>optk</b>     | integer   | Number of clusters                        |

**Returns:** Data frame with eigenvector columns, **Cluster**, and **Sample.ID**.

**Source:** ANF.R (line ~315)

### 6.3.3 extract\_eigenspace

```
extract_eigenspace(affinity, K, type = 3)
```

**Description:** Extracts the spectral embedding (eigenvectors) and cluster labels from an affinity matrix using wMKL discretization.

**Parameters:**

| Parameter       | Type    | Description                      |
|-----------------|---------|----------------------------------|
| <b>affinity</b> | matrix  | Square symmetric affinity matrix |
| <b>K</b>        | integer | Number of clusters               |
| <b>type</b>     | integer | Laplacian normalization type     |

**Returns:** List with **labels**, **U** (eigenvector matrix), and **eigDiscrete**.

**Source:** wMKL.R (line ~340)

## 6.4 Statistical Analysis Functions

### 6.4.1 unbiased.cv.test

```
unbiased.cv.test(x, string, digits = 3)
```

**Description:** Computes the bias-corrected Cramér’s V (or Phi coefficient for 2×2 tables) as a measure of association between two categorical variables.

**Parameters:**

| Parameter     | Type      | Description                          |
|---------------|-----------|--------------------------------------|
| <b>x</b>      | table     | Contingency table                    |
| <b>string</b> | character | Description for output labeling      |
| <b>digits</b> | integer   | Decimal places to round (default: 3) |

**Returns:** List with `text` (formatted result) and `value` (numeric Cramér’s V).

**Source:** Found in 18 algorithm scripts including: `ab-SNF.R`, `ANF.R`, `CIMLR.R`, `COCA.R`, `iClusterBayes.R`, `KLIC.R`, `LRAcluster.R`, `MDICC.R`, `MFA.R`, `MOFA.R`, `MONET.R`, `MSNE.R`, `NEMO.R`, `RWR-F.R`, `RWR-NF.R`, `SNF.R`, `Spectrum.R`, `wMKL.R`

---

## 6.5 Graph/Network Analysis Functions

### 6.5.1 `determine_edge_support`

```
determine_edge_support(edge_weights)
```

**Description:** Identifies which data modalities primarily support a given edge in a fused affinity network, based on relative edge weights.

**Parameters:**

| Parameter                 | Type    | Description                                       |
|---------------------------|---------|---------------------------------------------------|
| <code>edge_weights</code> | numeric | Vector of edge weights from each modality network |

**Returns:** Integer vector of indices indicating supporting modalities.

**Details:** Uses a 10% threshold to distinguish dominant modalities. Single index if one modality dominates (>10% higher), multiple indices for similar contributions, all indices if within 10% of maximum.

**Source:** `ab-SNF.R`, `ANF.R`, `MDICC.R`, `RWR-F.R`, `RWR-NF.R`, `SNF.R`

---

## 6.6 MONET-Specific Functions

### 6.6.1 `offset_correlation`

```
offset_correlation(cor.matrix, offset)
```

**Description:** Transforms a correlation matrix by subtracting the global mean and an additional offset, then setting diagonal to zero.

**Parameters:**

| Parameter               | Type    | Description                          |
|-------------------------|---------|--------------------------------------|
| <code>cor.matrix</code> | matrix  | Square correlation matrix            |
| <code>offset</code>     | numeric | Offset value to subtract (e.g., 0.2) |

**Returns:** Transformed correlation matrix with diagonal = 0.

**Source:** `MONET.R` (line ~106)

### 6.6.2 edge\_absolute\_mean

```
edge_absolute_mean(nx_graph)
```

**Description:** Calculates the mean of absolute edge weights in a NetworkX graph object.

**Parameters:**

| Parameter             | Type     | Description                                           |
|-----------------------|----------|-------------------------------------------------------|
| <code>nx_graph</code> | NetworkX | Reticulate-wrapped NetworkX graph with weighted edges |

**Returns:** Numeric scalar, mean absolute weight.

**Source:** MONET.R (line ~450)

---

### 6.6.3 get\_mod\_edge\_absmean

```
get_mod_edge_absmean(mod, omic_name)
```

**Description:** Computes the mean absolute edge weight within a MONET module for a specific omic network.

**Parameters:**

| Parameter              | Type      | Description                       |
|------------------------|-----------|-----------------------------------|
| <code>mod</code>       | object    | MONET module object               |
| <code>omic_name</code> | character | Name of the omic (e.g., "RNAseq") |

**Returns:** Numeric scalar, mean absolute weight; NA if module has <2 members.

**Source:** MONET.R (line ~464)

---

## 6.7 COCA-Specific Functions

### 6.7.1 hclust\_gap

```
hclust_gap(x, k)
```

**Description:** Wrapper function for `cluster::clusGap` that returns precomputed hierarchical clustering results for a given k.

**Parameters:**

| Parameter      | Type    | Description                                       |
|----------------|---------|---------------------------------------------------|
| <code>x</code> | matrix  | Data matrix (gap statistic framework requirement) |
| <code>k</code> | integer | Number of clusters to return                      |

**Returns:** Clustering result from `list_of_k[[paste0("k = ", k)]]`.

**Source:** COCA.R (line ~136)

---

## 6.8 PAMOGK Preprocessing Functions

### 6.8.1 `clean_rownames`

```
clean_rownames(df)
```

**Description:** Removes version number suffixes (e.g., “.1”, “.2”) from row names.

**Source:** PAMOGK\_prep.R (line ~15)

---

### 6.8.2 `remove_na_rownames`

```
remove_na_rownames(df)
```

**Description:** Filters out rows that have NA as their row name, with reporting.

**Source:** PAMOGK\_prep.R (line ~30)

---

### 6.8.3 `get_dimensions`

```
get_dimensions(x)
```

**Description:** Returns formatted string describing dimensions or “NULL”.

**Source:** PAMOGK\_prep.R (line ~56)

---

### 6.8.4 `fix_uniprot_row_names`

```
fix_uniprot_row_names(df)
```

**Description:** Converts row names stored as R vector strings to semicolon-separated format.

**Source:** PAMOGK\_prep.R (line ~88)

---

### 6.8.5 filter\_uniprot\_ids

```
filter_uniprot_ids(up_input, PAMOGK_UP_IDs)
```

**Description:** Filters data frames to retain only rows with UniProt IDs that exist in PAMOGK's database.

**Parameters:**

| Parameter     | Type      | Description                                       |
|---------------|-----------|---------------------------------------------------|
| up_input      | list      | List of data frames with UniProt IDs as row names |
| PAMOGK_UP_IDs | character | Vector of valid UniProt IDs from PAMOGK           |

**Returns:** List of filtered data frames with duplicates resolved by sum (SNPs) or MAD (continuous).

**Source:** PAMOGK\_prep.R (line ~142)

---

## 6.9 iClusterBayes Tuning Functions

### 6.9.1 prepare\_global\_data

```
prepare_global_data(data_list, K_values, var_name)
```

**Description:** Prepares a tidy data frame of Z<sub>ar</sub> values across K values for density plotting.

**Source:** iCB\_inspect\_HPC\_tune\_results.R (line ~28)

---

### 6.9.2 prepare\_modality\_data

```
prepare_modality_data(data_list, K_values, var_name)
```

**Description:** Prepares a tidy data frame of Beta/Gamma acceptance rates by modality and K.

**Source:** iCB\_inspect\_HPC\_tune\_results.R (line ~38)

---

### 6.9.3 get\_summary\_statistics\_text

```
get_summary_statistics_text(fit_list, K_values)
```

**Description:** Generates formatted text output summarizing acceptance rate statistics.

**Source:** iCB\_inspect\_HPC\_tune\_results.R (line ~50)

---

### 6.9.4 create\_global\_plot

```
create_global_plot(data, title)
```

**Description:** Creates a ggplot2 density plot of Z.ar values colored by K.

**Source:** iCB\_inspect\_HPC\_tune\_results.R (line ~81)

---

### 6.9.5 create\_modality\_plot

```
create_modality_plot(data, K, title)
```

**Description:** Creates a ggplot2 density plot of Beta/Gamma acceptance rates by modality.

**Source:** iCB\_inspect\_HPC\_tune\_results.R (line ~93)

---

### 6.9.6 compute\_robust\_metrics\_and\_score

```
compute_robust_metrics_and_score(tune_results, K_values)
```

**Description:** Computes median acceptance rates and composite score based on distance from optimal rate (0.234).

**Returns:** List with `per_K_metrics` and `final_score` (lower is better).

**Reference:** Roberts et al. (1997). Weak convergence and optimal scaling of random walk Metropolis algorithms.

**Source:** iCB\_inspect\_HPC\_tune\_results.R (line ~110)

---

### 6.9.7 penalize\_acceptance

```
penalize_acceptance(acc_rate, low_thres = 0.1, high_thres = 0.9, penalty = 0.05)
```

**Description:** Returns a penalty if acceptance rate falls outside acceptable bounds.

**Parameters:**

| Parameter               | Type    | Description                              |
|-------------------------|---------|------------------------------------------|
| <code>acc_rate</code>   | numeric | Acceptance rate (0-1)                    |
| <code>low_thres</code>  | numeric | Lower threshold (default: 0.1)           |
| <code>high_thres</code> | numeric | Upper threshold (default: 0.9)           |
| <code>penalty</code>    | numeric | Penalty if out of bounds (default: 0.05) |

**Returns:** 0 if within bounds, `penalty` otherwise.

**Source:** `iCB_inspect_HPC_tune_results.R` (line ~424)

## 6.10 HPC Folder Functions

### 6.10.1 LRAcluster\_HPC Functions (`full_LRAcluster_source.R`)

Functions implementing the LRAcluster algorithm for multi-omics integration with support for binary, gaussian, and poisson data types.

```
LRAcluster(data, types, dimension = 2, names = as.character(1:length(data)))
```

**6.10.1.1 LRAcluster (Main Function) Description:** Low-Rank Approximation clustering for integrating multiple data types. Performs dimension reduction while accounting for different data distributions.

**Parameters:**

| Parameter              | Type      | Description                                            |
|------------------------|-----------|--------------------------------------------------------|
| <code>data</code>      | list      | List of data matrices (features $\times$ samples)      |
| <code>types</code>     | character | Vector of data types (“binary”, “gaussian”, “poisson”) |
| <code>dimension</code> | integer   | Target dimension (default: 2)                          |
| <code>names</code>     | character | Names for each data type                               |

**Returns:** List with `coordinate` (sample coordinates) and `potential` (explained likelihood ratio).

**Source:** `LRAcluster_HPC/full_LRAcluster_source.R` (line 44)



```
nuclear_approximation(mat, dimension)
```

**6.10.1.2 nuclear\_approximation Description:** Computes low-rank matrix approximation using SVD with soft-thresholding on singular values.

**Parameters:**

| Parameter | Type    | Description  |
|-----------|---------|--------------|
| mat       | matrix  | Input matrix |
| dimension | integer | Target rank  |

**Returns:** Low-rank approximation of input matrix.

**Source:** LRAcluster\_HPC/full\_LRAcluster\_source.R (line 580)

### 6.10.1.3 Gaussian Data Functions

| Function           | Line | Description                                |
|--------------------|------|--------------------------------------------|
| check.gaussian.row | 219  | Validate gaussian row (non-all-NA)         |
| check.gaussian     | 230  | Clean gaussian matrix, remove invalid rows |
| base.gaussian.row  | 248  | Compute row mean for baseline              |
| base.gaussian      | 254  | Compute baseline matrix (row means)        |
| update.gaussian    | 262  | Gradient update step                       |
| stop.gaussian      | 273  | Convergence criterion                      |
| LL.gaussian        | 285  | Log-likelihood computation                 |
| LLmax.gaussian     | 294  | Maximum log-likelihood (0)                 |
| LLmin.gaussian     | 299  | Minimum (null) log-likelihood              |
| gaussian_base      | 307  | Standalone gaussian fitting                |

### 6.10.1.4 Poisson Data Functions

| Function          | Line | Description                            |
|-------------------|------|----------------------------------------|
| check.poisson.row | 333  | Validate poisson row (non-negative)    |
| check.poisson     | 353  | Clean poisson matrix, add +1 to counts |
| base.poisson.row  | 370  | Compute log-mean for baseline          |
| base.poisson      | 378  | Compute baseline matrix (log-means)    |
| update.poisson    | 386  | Gradient update step                   |
| stop.poisson      | 397  | Convergence criterion                  |
| LL.poisson        | 407  | Log-likelihood computation             |
| LLmax.poisson     | 415  | Maximum log-likelihood                 |
| LLmin.poisson     | 422  | Minimum (null) log-likelihood          |
| poisson_type_base | 429  | Standalone poisson fitting             |

### 6.10.1.5 Binary Data Functions

| Function                      | Line | Description                               |
|-------------------------------|------|-------------------------------------------|
| <code>check.binary.row</code> | 454  | Validate binary row (has variation)       |
| <code>check.binary</code>     | 473  | Clean binary matrix, remove constant rows |
| <code>base.binary.row</code>  | 488  | Compute log-odds for baseline             |
| <code>base.binary</code>      | 496  | Compute baseline matrix (log-odds)        |
| <code>update.binary</code>    | 504  | Logistic gradient update                  |
| <code>stop.binary</code>      | 518  | Convergence criterion                     |
| <code>LL.binary</code>        | 532  | Log-likelihood computation                |
| <code>LLmax.binary</code>     | 543  | Maximum log-likelihood (0)                |
| <code>LLmin.binary</code>     | 548  | Minimum (null) log-likelihood             |
| <code>binary_type_base</code> | 558  | Standalone binary fitting                 |

### 6.10.1.6 Utility Functions

| Function                          | Line | Description                      |
|-----------------------------------|------|----------------------------------|
| <code>check.matrix.element</code> | 1    | Check if element is NOT a matrix |
| <code>ncol.element</code>         | 13   | Wrapper for <code>ncol()</code>  |
| <code>nrow.element</code>         | 18   | Wrapper for <code>nrow()</code>  |
| <code>check</code>                | 23   | Type dispatcher for validation   |

## 6.10.2 RWR-F\_HPC Functions (RWR-F\_source.R)

Random Walk with Restart based network fusion for multi-omics integration.

```
RWR_fusion_neighbor(sim_list, iteration_max = 1000, gama = 0.7,  
                    neighbor_num = 10, alpha = 0.9, beta = 0.9)
```

**6.10.2.1 RWR\_fusion\_neighbor Description:** RWR-based similarity network fusion with neighbor-based transitions. Integrates multiple similarity networks into a single fused matrix.

**Parameters:**

| Parameter                  | Type    | Description                                |
|----------------------------|---------|--------------------------------------------|
| <code>sim_list</code>      | list    | List of similarity matrices                |
| <code>iteration_max</code> | integer | Max iterations (default: 1000)             |
| <code>gama</code>          | numeric | Restart probability (default: 0.7)         |
| <code>neighbor_num</code>  | integer | Number of neighbors (default: 10)          |
| <code>alpha</code>         | numeric | Self-network weight (default: 0.9)         |
| <code>beta</code>          | numeric | Self-transition probability (default: 0.9) |

**Returns:** Fused similarity matrix (normalized, symmetric).

**Note:** Requires `foreach` and `doParallel` packages for parallel execution.

**Source:** `RWR-F_HPC/RWR-F_source.R` (line 1)

---

```
RWR_fusion(sim_list, iteration_max = 1000, gama = 0.7)
```

**6.10.2.2 RWR\_fusion Description:** Standard RWR-based network fusion (simpler variant without neighbor-based transitions).

**Parameters:**

| Parameter                  | Type    | Description                        |
|----------------------------|---------|------------------------------------|
| <code>sim_list</code>      | list    | List of similarity matrices        |
| <code>iteration_max</code> | integer | Max iterations (default: 1000)     |
| <code>gama</code>          | numeric | Restart probability (default: 0.7) |

**Returns:** Fused similarity matrix (normalized, symmetric).

**Source:** `RWR-F_HPC/RWR-F_source.R` (line 176)

---

## 7 Scripts/Consensus Functions

Functions for consensus clustering analysis and post-processing.

### 7.1 calc\_ari

```
calc_ari(set1, set2, universe)
```

**Description:** Computes Adjusted Rand Index (ARI) between two sets of pathways by converting them to binary membership vectors.

**Parameters:**

| Parameter             | Type      | Description               |
|-----------------------|-----------|---------------------------|
| <code>set1</code>     | character | Pathway IDs in first set  |
| <code>set2</code>     | character | Pathway IDs in second set |
| <code>universe</code> | character | All possible pathway IDs  |

**Returns:** Numeric ARI value between -1 and 1.

**Source:** `Post_consensus.R` (line 169), `Post_consensus_more_than_2_clusters.R` (line 157)

---

## 8 Scripts/MOVICS Functions

Functions for MOVICS baseline analysis and output inspection.

### 8.1 `check_missing_values_with_indices`

```
check_missing_values_with_indices(input)
```

**Description:** Examines each matrix in a list for missing values, returning both count and row indices.

**Parameters:**

| Parameter          | Type | Description               |
|--------------------|------|---------------------------|
| <code>input</code> | list | List of matrices to check |

**Returns:** Named list with `na_count` and `missing_indices` for each matrix.

**Source:** `MOVICS_baseline.R` (line 227)

---

### 8.2 `preprocessText`

```
preprocessText(textVector, removeStopwords = FALSE)
```

**Description:** Converts text strings to term-document matrix and extracts term frequencies for wordclouds.

**Parameters:**

| Parameter                    | Type      | Description                               |
|------------------------------|-----------|-------------------------------------------|
| <code>textVector</code>      | character | Text strings to process                   |
| <code>removeStopwords</code> | logical   | Remove English stopwords (default: FALSE) |

**Returns:** Data frame with `term` and `freq` columns.

**Source:** `MOVICS_consensus_output_inspection.R` (line 580), `Comparisons.R` (line 505)

---

### 8.3 `overlap_coefficient`

```
overlap_coefficient(set1, set2)
```

**Description:** Computes Szymkiewicz-Simpson overlap coefficient:  $\text{intersection} / \min(\text{set sizes})$ .

**Parameters:**

| Parameter         | Type   | Description |
|-------------------|--------|-------------|
| <code>set1</code> | vector | First set   |
| <code>set2</code> | vector | Second set  |

**Returns:** Numeric value 0-1. Returns 0 if either set is empty.

**Source:** `MOVICS_consensus_output_inspection.R` (line 644), `Comparisons.R` (line 383)

---

## 8.4 calculate\_similarities

```
calculate_similarities(algorithms, pathway_column, pathway_list)
```

**Description:** Computes pairwise similarity matrix between algorithms using overlap coefficient.

**Parameters:**

| Parameter                   | Type       | Description                                   |
|-----------------------------|------------|-----------------------------------------------|
| <code>algorithms</code>     | character  | Algorithm names                               |
| <code>pathway_column</code> | unquoted   | Column containing pathway lists               |
| <code>pathway_list</code>   | data.frame | Data frame with Algorithm and pathway columns |

**Returns:** Square numeric similarity matrix.

**Source:** `MOVICS_consensus_output_inspection.R` (line 649)

---

## 9 Scripts/method\_comparisons/Comparisons.R Functions

Functions for method comparison analysis.

### 9.1 calculate\_pathway\_similarity\_matrix

```
calculate_pathway_similarity_matrix(agg_list)
```

**Description:** Computes pairwise similarity matrix from a named list of pathway sets.

**Parameters:**

| Parameter             | Type | Description                   |
|-----------------------|------|-------------------------------|
| <code>agg_list</code> | list | Named list of pathway vectors |

**Returns:** Square numeric similarity matrix.

**Source:** Comparisons.R (line 389)

## 9.2 calculate\_ari\_pathway

```
calculate_ari_pathway(agg_path1, agg_path2, all_pathways)
```

**Description:** Computes ARI between two pathway sets over a common universe.

**Parameters:**

| Parameter                 | Type      | Description              |
|---------------------------|-----------|--------------------------|
| <code>agg_path1</code>    | character | First pathway set        |
| <code>agg_path2</code>    | character | Second pathway set       |
| <code>all_pathways</code> | character | Universe of all pathways |

**Returns:** Numeric ARI value (-1 to 1).

**Source:** Comparisons.R (line 544)

## 9.3 cell\_fun (inline)

```
cell_fun(j, i, x, y, width, height, fill)
```

**Description:** Custom cell drawing function for ComplexHeatmap with dual-color schemes.

**Details:** Handles diagonal cells (white), NA values, and applies different color schemes for cluster vs pathway ARI values. Requires external variables: `M`, `type_mat`, `col_fun_cluster`, `col_fun_pathway`.

**Source:** Comparisons.R (line 677)

## 9.4 pca\_from\_ari\_matrix

```
pca_from_ari_matrix(sim_matrix = NULL, clust_res = NULL, cluster_colors = NULL,
                     output_path = NULL, pointsize = NULL, geom_label_size = NULL)
```

**Description:** Performs kernel PCA on an ARI matrix and creates a 2D visualization with category coloring.

**Parameters:**

| Parameter                    | Type         | Description                             |
|------------------------------|--------------|-----------------------------------------|
| <code>sim_matrix</code>      | matrix       | Square ARI similarity matrix            |
| <code>clust_res</code>       | data.frame   | Algorithm metadata with Category column |
| <code>cluster_colors</code>  | named vector | Colors for each category                |
| <code>output_path</code>     | character    | Output path (unused)                    |
| <code>pointsize</code>       | numeric      | Point size (default: 2)                 |
| <code>geom_label_size</code> | numeric      | Label size (default: 2)                 |

**Returns:** ggplot object with kernel PCA projection.

**Source:** Comparisons.R (line 1157)

---

## 10 Top-level Scripts Functions

Functions found in top-level scripts within the `Scripts/` folder.

---

### 10.1 `check_duplicate_rownames`

```
check_duplicate_rownames(data_object)
```

**Description:** Checks for duplicate rownames in a list of matrices. Identifies which matrices have duplicated rownames and returns a summary.

**Parameters:**

| Parameter                | Type | Description                                            |
|--------------------------|------|--------------------------------------------------------|
| <code>data_object</code> | list | Named list of matrices to check for duplicate rownames |

**Returns:** Named character vector containing duplicated rownames for matrices that have duplicates, or “No duplicates found” messages.

**Details:** Also found in `RNAseq_clean_counts.R` with parameter name `rna_object`.

**Source:** `Download_TCGA_data.R` (line 497), `RNAseq_clean_counts.R` (line 14)

---

### 10.2 `filter_most_variant_rows`

```
filter_most_variant_rows(data, data_row_names)
```

**Description:** Handles duplicate rownames by keeping only the most variant rows. For genes with multiple probe mappings, retains the probe/row with highest variance.

**Parameters:**

| Parameter                   | Type      | Description                                   |
|-----------------------------|-----------|-----------------------------------------------|
| <code>data</code>           | matrix    | Data matrix with numeric values               |
| <code>data_row_names</code> | character | Vector of rownames (possibly with duplicates) |

**Returns:** Matrix with unique rownames, keeping only the most variant row for each duplicated gene.

**Dependencies:** Requires `matrixStats::rowVars()` for variance calculation.

**Source:** `Download_TCGA_data.R` (line 528), `RNAseq_clean_counts.R` (line 43)

---



## 11 Alphabetical Index

| Function                            | Source File                                                       |
|-------------------------------------|-------------------------------------------------------------------|
| as.sunburstDF                       | custom_functions.R                                                |
| base.binary                         | LRACluster_HPC/full_LRACluster_source.R                           |
| base.binary.row                     | LRACluster_HPC/full_LRACluster_source.R                           |
| base.gaussian                       | LRACluster_HPC/full_LRACluster_source.R                           |
| base.gaussian.row                   | LRACluster_HPC/full_LRACluster_source.R                           |
| base.poisson                        | LRACluster_HPC/full_LRACluster_source.R                           |
| base.poisson.row                    | LRACluster_HPC/full_LRACluster_source.R                           |
| binary_type_base                    | LRACluster_HPC/full_LRACluster_source.R                           |
| calc_ari                            | Post_consensus.R,<br>Post_consensus_more_than_2_clusters.R        |
| calculate_ari_index                 | custom_functions.R                                                |
| calculate_ari_pathway               | Comparisons.R                                                     |
| calculate_nmi_index                 | custom_functions.R                                                |
| calculate_pathway_similarity_matrix | Comparisons.R                                                     |
| calculate_S                         | custom_functions.R                                                |
| calculate_similarities              | MOVICS_consensus_output_inspection.R                              |
| cell_fun                            | Comparisons.R                                                     |
| check                               | LRACluster_HPC/full_LRACluster_source.R                           |
| check.binary                        | LRACluster_HPC/full_LRACluster_source.R                           |
| check.binary.row                    | LRACluster_HPC/full_LRACluster_source.R                           |
| check.gaussian                      | LRACluster_HPC/full_LRACluster_source.R                           |
| check.gaussian.row                  | LRACluster_HPC/full_LRACluster_source.R                           |
| check.matrix.element                | LRACluster_HPC/full_LRACluster_source.R                           |
| check.poisson                       | LRACluster_HPC/full_LRACluster_source.R                           |
| check.poisson.row                   | LRACluster_HPC/full_LRACluster_source.R                           |
| check_duplicate_rownames            | Download_TCGA_data.R, RNAseq_clean_counts.R                       |
| check_missing_values_with_indices   | MOVICS_baseline.R, Download_TCGA_data.R,<br>RNAseq_clean_counts.R |
| CIMLR_mod                           | custom_functions.R                                                |
| clean_rownames                      | PAMOGK_prep.R                                                     |
| compute_matrix_similarity           | ab-SNF.R, ANF.R, RWR-F.R, RWR-NF.R, SNF.R,<br>Spectrum.R          |
| compute_robust_metrics_and_score    | iCB_inspect_HPC_tune_results.R                                    |
| create_global_plot                  | iCB_inspect_HPC_tune_results.R                                    |
| create_modality_plot                | iCB_inspect_HPC_tune_results.R                                    |
| determine_edge_support              | ab-SNF.R, ANF.R, MDICC.R, RWR-F.R, RWR-NF.R,<br>SNF.R             |
| edge_absolute_mean                  | MONET.R                                                           |
| extract_eigenspace                  | wMKL.R                                                            |
| fetch_citation                      | custom_functions.R                                                |
| fetch_in_a_nutshell                 | custom_functions.R                                                |
| filter_most_variant_rows            | Download_TCGA_data.R, RNAseq_clean_counts.R                       |
| filter_uniprot_ids                  | PAMOGK_prep.R                                                     |
| fix_uniprot_row_names               | PAMOGK_prep.R                                                     |
| frobenius_norm                      | custom_functions.R                                                |
| fuzzy_term_clustering_fast          | fast_pathfindR_hclust.R                                           |
| gaussian_base                       | LRACluster_HPC/full_LRACluster_source.R                           |
| get_dimensions                      | PAMOGK_prep.R                                                     |
| get_mod_edge_absmean                | MONET.R                                                           |

| Function                          | Source File                                            |
|-----------------------------------|--------------------------------------------------------|
| get_summary_statistics_text       | iCB_inspect_HPC_tune_results.R                         |
| getMoHeatmap_prenorm_quart        | custom_functions.R                                     |
| getMoHeatmap_single_algorithm     | modified_MOVICS_functions.R                            |
| getMoHeatmap_single_algorithm2    | modified_MOVICS_functions.R                            |
| getSilhouette_ggplot              | modified_MOVICS_functions.R                            |
| hclust_gap                        | COCA.R                                                 |
| hierarchical_term_clustering_fast | fast_pathfindR_hclust.R                                |
| kernfinder_local_mod              | Spectrum_functions.R                                   |
| kernfinder_mine_mod               | Spectrum_functions.R                                   |
| linear_quantile_normalize         | custom_functions.R                                     |
| LL.binary                         | LRAcluster_HPC/full_LRAcluster_source.R                |
| LL.gaussian                       | LRAcluster_HPC/full_LRAcluster_source.R                |
| LL.poisson                        | LRAcluster_HPC/full_LRAcluster_source.R                |
| LLmax.binary                      | LRAcluster_HPC/full_LRAcluster_source.R                |
| LLmax.gaussian                    | LRAcluster_HPC/full_LRAcluster_source.R                |
| LLmax.poisson                     | LRAcluster_HPC/full_LRAcluster_source.R                |
| LLmin.binary                      | LRAcluster_HPC/full_LRAcluster_source.R                |
| LLmin.gaussian                    | LRAcluster_HPC/full_LRAcluster_source.R                |
| LLmin.poisson                     | LRAcluster_HPC/full_LRAcluster_source.R                |
| LRAcluster                        | LRAcluster_HPC/full_LRAcluster_source.R                |
| mad_based_weights                 | ab-SNF.R, wMKL.R                                       |
| mds_from_original_matrix          | custom_functions.R                                     |
| MOVICS_jaccard_index              | custom_functions.R                                     |
| multiple.kernel.cimlr_mod         | custom_functions.R                                     |
| multiple.kernel.cimlr.weight_mod  | custom_functions.R                                     |
| ncol.element                      | LRAcluster_HPC/full_LRAcluster_source.R                |
| nemo.affinity.graph_mod           | custom_functions.R                                     |
| normalize_affinity_matrix         | custom_functions.R                                     |
| nrow.element                      | LRAcluster_HPC/full_LRAcluster_source.R                |
| ntp_mod                           | modified_MOVICS_functions.R                            |
| nuclear_approximation             | LRAcluster_HPC/full_LRAcluster_source.R                |
| offset_correlation                | MONET.R                                                |
| overlap_coefficient               | MOVICS_consensus_output_inspection.R,<br>Comparisons.R |
| pca_from_ari_matrix               | Comparisons.R                                          |
| pca_from_original_matrix          | custom_functions.R                                     |
| pca_from_sim_matrix               | custom_functions.R                                     |
| pearson_correlation               | custom_functions.R                                     |
| penalize_acceptance               | iCB_inspect_HPC_tune_results.R                         |
| plot_MOFA_cumul_var_exp           | custom_functions.R                                     |
| plot_MOFA_var_exp                 | custom_functions.R                                     |
| plot_pathway_heatmaps             | modified_MOVICS_functions.R                            |
| poisson_type_base                 | LRAcluster_HPC/full_LRAcluster_source.R                |
| prepare_global_data               | iCB_inspect_HPC_tune_results.R                         |
| prepare_gsea_output_for_hclust    | modified_MOVICS_functions.R                            |
| prepare_modality_data             | iCB_inspect_HPC_tune_results.R                         |
| preprocessText                    | MOVICS_consensus_output_inspection.R,<br>Comparisons.R |
| rankFeaturesByNMI_parallely       | custom_functions.R                                     |
| rankFeaturesByNMI_parallely_ANF   | custom_functions.R                                     |
| rbfkernel_b_mod                   | Spectrum_functions.R                                   |
| remove_na_rownames                | PAMOGK_prep.R                                          |

| Function                             | Source File                             |
|--------------------------------------|-----------------------------------------|
| reshape_Pearson_for_tests            | custom_functions.R                      |
| runDEA_mod                           | modified_MOVICS_functions.R             |
| runGSEA_mod                          | modified_MOVICS_functions.R             |
| runGSEA_mod_4.4                      | modified_MOVICS_functions.R             |
| runGSEA_mod_4.4_single_algorithm     | modified_MOVICS_functions.R             |
| runGSVA_mod_4.4                      | modified_MOVICS_functions.R             |
| runGSVA_mod_4.4_single_algorithm     | modified_MOVICS_functions.R             |
| runKappa_single_algorithm            | modified_MOVICS_functions.R             |
| runMarker_mod_4.4                    | modified_MOVICS_functions.R             |
| runMarker_single_algorithm           | modified_MOVICS_functions.R             |
| runMarker_single_algorithm_no_export | modified_MOVICS_functions.R             |
| runNTP_mod                           | modified_MOVICS_functions.R             |
| runPAM_single_algorithm              | modified_MOVICS_functions.R             |
| RWR_fusion                           | RWR-F_HPC/RWR-F_source.R                |
| RWR_fusion_neighbor                  | RWR-F_HPC/RWR-F_source.R                |
| spectral_clustering_eig              | ANF.R                                   |
| spectralClustering_eig               | ab-SNF.R, SNF.R, NEMO.R                 |
| Spectrum_bin_and_par                 | Spectrum_functions.R                    |
| standardize_rows                     | custom_functions.R                      |
| stop.binary                          | LRACluster_HPC/full_LRACluster_source.R |
| stop.gaussian                        | LRACluster_HPC/full_LRACluster_source.R |
| stop.poisson                         | LRACluster_HPC/full_LRACluster_source.R |
| subHeatmap_mod                       | modified_MOVICS_functions.R             |
| TCGAanalyze_survival_custom3         | custom_functions.R                      |
| twoclassdeseq2_mod                   | modified_MOVICS_functions.R             |
| twoclassedger_mod                    | modified_MOVICS_functions.R             |
| twoclasslimma_mod                    | modified_MOVICS_functions.R             |
| unbiased.cv.test                     | 18 algorithm scripts                    |
| update.binary                        | LRACluster_HPC/full_LRACluster_source.R |
| update.gaussian                      | LRACluster_HPC/full_LRACluster_source.R |
| update.poisson                       | LRACluster_HPC/full_LRACluster_source.R |
| weighted_euclidean_dist              | ab-SNF.R                                |
| weighted_hamming_fraction            | ab-SNF.R                                |
| weighted_mutations                   | wMKL.R                                  |

---

*Documentation auto-generated for MultiOmicsSurvey project.*